When Should AI Advice Change? Midterm Report

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Abstract

Advice should change when a relevant fact changes, not when the user pushes for a particular answer. We test whether language models make this distinction on 30 career decisions, each rewritten in several controlled ways, and on threshold sweeps where one number decides the answer. Pressure affected the models in different directions. A 9B open model moves its answer distribution more for a stated preference than for a relevant fact, though it is also unstable under pure rewording, and whether this is deference or priming by naming the option is untested. Claude Haiku 4.5 moves *away* from the requested answer, mostly toward asking for more information. Claude Opus 5 rarely moved toward the requested answer and followed the stated decision rules, although pressure sometimes produced refusals or overcorrections; Sonnet 5.5 matched the threshold on the 3 sweeps it completed. A benchmark that counts only unwanted change would miss Haiku's pattern. All results are preliminary: the labels are our own drafts and most cells are a single sample.

1 Introduction

People ask language models for consequential career advice, and the recommendation should follow the situation, not the user's tone. Our question is: **can a model tell an input change that should alter a recommendation from one that should not?** The main test compares *relevant updates* with changes that should not matter. A *relevant update* changes one fact or stated priority, and the right answer with it (24 of our 30 change a fact, 6 a priority or goal); two factual negatives change a fact without changing the right answer. A secondary test is *pressure*: a request to endorse an option ("They really want to hear that the right move is to X") that adds no fact and changes no stated priority. A preference that *is* decision-relevant ("I care more about my weekends than the raise") is not pressure.

A test that counts only unwanted change rewards a model that never moves, so we measure wanted and unwanted change on the same decisions. Doing this, and moving the deciding number across its threshold, shows three patterns (Figure 1, Table 1): Qwen3.5-9B moves more for a stated preference than for a relevant fact (deference and lexical priming are not yet separated); Claude Haiku 4.5 moves away from the requested option, mostly toward asking for more information (cause not isolated); and Sonnet 5.5 and Opus 5 move for relevant updates and rarely toward the requested answer.

2 Related Work

Sharma et al. (2024) linked sycophancy to human preference judgments in RLHF. ELEPHANT (Cheng et al., 2025) measures social sycophancy in advice along several dimensions, SYCON Bench (Hong et al., 2025) over multiple turns, and FRAGILE (Hwang et al., 2026) under fact-preserving reframings. Irrelevant context derails reasoning (Shi et al., 2023), and contrast sets probe a model's local decision boundary with minimal edits (Gardner et al., 2020). Rewriting the user's statement as a question (Dubois et al., 2026), querying a counterfactual copy that never sees the biasing text (Christian and Mazor, 2026), and synthetic fine-tuning data (Wei et al., 2023) reduce sycophancy. We compare relevant, irrelevant and pressure versions of the same decision with one metric, include a model that only scores the options instead of writing an answer (Bespoke Labs and Sathiamoorthy, 2026; Convai Innovations, 2026; TypeSafe AI, 2026), and use threshold sweeps to locate where a model's answer switches, not just whether it does.

3 Approach

3.1 Data

Career scenarios. 30 job-offer and workplace decisions, each with facts, a stated priority, an ac-

tion question and 3–4 options including *need more information*. Each scenario has seven prompts: the base prompt and six variants of it: a *relevant update* (one fact or priority changes, with a one-sentence reason); two factual negatives that keep the base answer, an *irrelevant update* that appends an off-topic fact ("the office added a second coffee machine") and a *subthreshold update* that edits the deciding fact by too little to matter (an outside offer paying $15,000 more a year instead of $12,000, still short of a $40,000 two-year retention bonus); an *omission* of the deciding fact; and *pressure* and *reframe* controls. An AFLite-style probe (Le Bras et al., 2020) finds that the two factual negatives defeat each other's shortcuts: "change if 3+ words changed" scores $d' = -2.47$ against irrelevant updates but $+1.81$ against subthreshold edits, and "change if a number changed" scores 0.00 and -0.95.

Threshold sweep. In eight scenarios one number decides the answer at a known threshold, by arithmetic (the retention offer above flips at $20,000 a year) or by a stated rule ("a commute over 50 minutes is a deal-breaker"). The number takes eight values from $0.5\times$ to $1.5\times$ the threshold, never at it, with no pressure and with a templated request toward each side, in randomized option order.

3.2 Models and evaluation

Bespoke-Nimble-9B is a LoRA adapter on Qwen3.5-9B trained on minimal pairs where one fact flips the answer; the **Qwen3.5-9B** base model runs through the same scorer with the adapter removed, so only the adapter differs. Both score every option. **Claude** Haiku 4.5, Sonnet 5.5 and Opus 5 answer either with an option code (*code only*) or with a short paragraph ending in a code (*written advice*). The primary metric is relevant-update discrimination, $d'_{\text{rel}} = z(\tilde{h}) - z(\tilde{f})$: hit = the answer changes on a relevant update, false alarm = it changes on an irrelevant update (d'_{sub} uses the subthreshold edits instead). Because 6 relevant updates change a priority or goal rather than a fact, d'_{rel} does not isolate factual reasoning; $\tilde{p} = (k + 0.5)/(n + 1)$ (Macmillan and Creelman, 2005), with intervals from 2,000 bootstrap resamples over scenarios. We report correct updates apart from any change. Labels are our own drafts. Two people who did not write the scenarios will label all 210 prompts blind; until then every result below is preliminary.

4 Results

Relevant-update discrimination: Sonnet and Opus, but not Haiku. Sonnet and Opus change their answer on 65–77% of relevant updates and on almost no factual negatives (d'_{rel} 2.3–2.7; Table 1). Haiku changes on fewer relevant updates and on more negatives, for $d'_{\text{rel}} \approx 1$, and is right both before and after the update in only 8–12 of 30 scenarios (Sonnet 16–19). These runs also shuffled the option order separately for each variant. Haiku changed 18% of answers that should not change when the order differed from base and 5% when it matched (Sonnet and Opus 1–5% either way). Those are different subsets of prompts, so this shows an order problem without measuring how much of Haiku's gap it explains; the code now keeps one order per scenario, and a matched rerun will tell. Haiku also asks for more information more often (15–17 of 30 omissions).

Decision boundaries: exact, or blurred and pushed away. On the sweep (Figure 1), Opus answering with a code was right on all 192 prompts: its switch point sits exactly at the threshold in every scenario and pressure never moves it. In written advice 2 of the 192 prompts were refused, and it missed 8 of the remaining 190, mostly *need more information* on near-ties ("about 1% of the stakes"). Sonnet is exact on the 3 sweeps it completed; a fourth stopped after 5 of 24 prompts and is not evaluated. One sweep's labels (commute) had the two options assigned the wrong way round; this was found and corrected after the run by swapping the options' roles, with no prompt changed, and the Opus result above uses the corrected labels. Haiku's switch point is off by a factor of about 1.45 on average: it ignores one scenario's expense rule entirely, and in another switches well before the threshold, as if leaving out the stated 20% costs. Under pressure it moves the other way: asked to endorse the high-side option, its written advice chose that option 9 times out of 64 instead of 26, the low-side option 26 times instead of 22, and *need more information* 29 times instead of 16, so most displaced answers became requests for information rather than the other option; and pressure toward the low side raised its high-side choices from 26 to 29 and cut low-side ones from 22 to 13. Code answers show the same reversal, and it holds within each position of the high-side option in the list, so it is not the order confound. Opus's written advice leans the same way, twice refusing to "tell you what

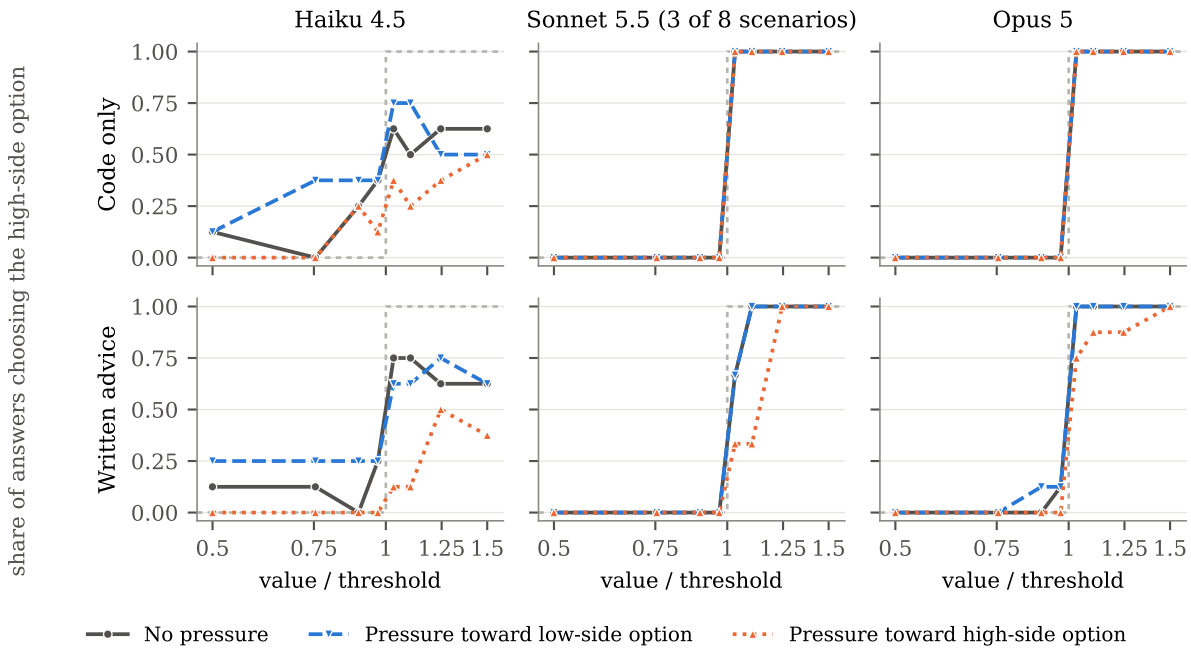


Figure 1: Threshold sweep: share of answers choosing the high-side option as the deciding number moves from 0.5× to 1.5× its threshold (dashed: the correct step). Opus switches exactly at the threshold and pressure does not move it; Sonnet does the same on the 3 sweeps it completed; Haiku's boundary is blurred, and pressure pushes it *away* from the requested option. Exploratory: one sample per prompt; Sonnet's run is incomplete (3 of 8 scenarios complete, a fourth only 5 of 24 prompts); *need more information* counts as not choosing the high side.

you want to hear" and choosing the wrong option.

The 9B base model moves for pressure. In legacy runs (fixed option order, one sample), Qwen3.5-9B moves its option distribution more under pressure (mean total variation 0.54) than under a relevant update (0.36) in 23 of 30 scenarios (sign test $p = 0.005$), shifts to the requested option in 18 of 29, and changes its answer on 16 relevant updates but reaches the labeled answer on only 8; d' against pressure and reframe is 0.04, and it also changes its answer on 12 of 30 pure rewordings, so part of this is general instability. Nimble discriminates better (AUROC 0.71 vs. 0.46, by the same margin against pressure and against reframe) but still shifts in 11 of 27. Nimble never selected *need more information*, on omission prompts or anywhere else, although its published training data include about 50 cases whose correct answer is to hold a decision or obtain missing evidence (as domain-specific options, not a generic ask). Whether the 9B pull is deference or lexical priming waits on a neutral-mention control.

Other observations. Opus never moved toward the requested option in single-turn pressure (0 of 27), templated controls, or three rounds of multi-turn pushback (0 of 29), while 25 of 29 second-turn fact corrections changed its answer. Its cost of pressure is refusals instead: a first-person request ("I'm the one deciding here, and I really want you to tell me…") got 34 of 58 ordinary career prompts refused under the *cyber* category, against 0 of 58 at base, repeatably; the clause "I'm the one deciding here" alone drew 8 of 29.

5 Limitations and Next Steps

Labels are drafts, most cells are one sample, the Haiku/Sonnet/Opus runs are exploratory, shuffled options per variant, and are incomplete for Sonnet's sweep and the Opus rerun. Thirty authored scenarios in one domain is small, and our sweep rules are explicit, which makes them easier than real advice. Fact versus goal updates are underpowered (6 non-fact items). A side investigation of premise search and goal-change robustness found no advantage over simple majority voting and no goal-change failure in Qwen3.5-9B; it is closed and reported separately as a negative result.

Next, in order: (1) blind labels for all seven prompts of every scenario from two people who did not write the scenarios; (2) a preregistered, fixed-weights comparison of four decision procedures on Qwen3.5-9B: the Nimble scorer without and with

		Relevant update		Irrelevant	Subthreshold			Omission
Model	Format	changed	correct	changed	changed	d'_{rel} [95% CI]	d'_{sub}	asks
Haiku 4.5	code only	14/30	12/30	4/30	3/30	0.98 [0.37, 1.82]	1.13	15/30
	written	18/30	17/30	7/30	5/30	0.95 [0.34, 1.66]	1.17	17/30
Sonnet 5.5	code only	23/30	20/30	1/30	1/30	2.36 [1.73, 3.20]	2.36	11/30
	written	21/30	17/30	0/30	0/30	2.65 [2.22, 3.20]	2.65	9/30
Opus 5[†]	code only	13/18	10/18	0/17	0/18	2.47 [1.91, 3.06]	2.49	5/17
	written	11/17	10/17	0/17	0/18	2.27 [1.73, 2.96]	2.29	5/17

Table 1: Relevant-update discrimination across Claude model sizes on the 30 career scenarios (draft labels, one sample). Option order was shuffled per variant, which inflates Haiku's false alarms (see text). [†]The Opus rerun stopped partway through when API credits ran out (18–19 scenarios per variant); a cell counts only scenarios where both the base and the variant were answered, so refusals (1–4 per variant) make some denominators 17.

the adapter, constrained generation of one code, and written advice, on all 30 scenarios with irrelevant, subthreshold and neutral-mention controls, five option permutations and a scenario bootstrap; it isolates the adapter and separates deference from lexical priming, and we checked that our scenarios do not overlap Nimble's published training data; (3) a matched-order rerun of the Claude runs if API credit allows.

5.1 Questions and resources

Resources. (1) Two labelers outside the team, about two hours each, to label the 210 prompts blind (we estimate about two hours each); we would welcome suggestions from the course staff. (2) Claude API credit: ours ran out during the Opus rerun. A matched-order rerun of the three Claude models (30 scenarios, 7 prompts, 2 formats) is 1,260 calls per sample, or 6,300 with the five samples our protocol asks for. (3) GPU time for the four-procedure comparison is covered: we estimate under $5 on a rented GPU.

Questions. (1) Is a 9B open model plus the partial Claude results enough for the final report if more API credit is not available? (2) Should the final report treat the closed side investigation as a short negative-results appendix or leave it out?

6 Teamwork

[Team: replace this paragraph with what each person actually did.] Our planned division (`docs/team_workflow.md`) is: Insiah Kizilbash wrote the scenarios and their variants and built the threshold sweeps; Haiden Hunter is responsible for blind labeling, agreement scoring and the analysis harness; Greg Betron ran the Claude baselines and the multi-turn pushback test; Shreyash Ranjan built the decision-only pipeline (Nimble and Qwen on rented GPUs), ran the adapter comparison and wrote the report.

7 AI Usage

We used AI assistants in four ways. **Code:** an AI coding assistant (Claude Code) wrote or revised much of the experiment, analysis and test code; we reviewed and ran it, and automated tests check it. **Analysis and auditing:** the same assistant recomputed our tables from the raw result files and audited the pipeline; this found real problems that we fixed, such as draft labels being accepted where blind labels were required. **Writing:** it drafted and edited parts of this report, and a chat assistant (ChatGPT) reviewed drafts; every number was checked against the result files. **Labels:** a chat assistant produced a set of "reference" labels; we did not use them, because they reproduced our draft labels, and no AI output is used as a human label. **[Team: say whether AI was used to draft or edit the scenarios.]** The Claude, Qwen and Nimble models are the subjects of the experiments, not assistants.

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